the battery system to be contained within a smaller space.

NiMH batteries do have some drawbacks such as lower charging efficiencies than other batteries. There is also an issue with self-discharge (up to 12.5 per cent per day under normal room temperature conditions) that is exacerbated when the batteries are in a high temperature environment. This makes NiMH batteries less ideal for hotter environments.

Zebra. The sodium or zebra battery uses a molten chloroaluminate-sodium (NaAlCl4) as the electrolyte. This chemistry is also occasionally referred to as hot salt. A relatively mature technology, the zebra battery has an energy density of 120Wh/kg and reasonable series resistance.

Since the batteries must be heated for use, cold weather does not strongly affect their operation, except increasing heating costs. Zebra batteries have been used in several EVs. These can last for a few thousand charge cycles and are non-toxic.

Downsides to the zebra battery include poor power density (< 300W/kg) and the requirement of having to heat the electrolyte to about 270°C (520°F), which wastes some energy and presents difficulties in long-term storage of charge.

Flow batteries. A flow battery, or redox flow battery (after reduction-oxidation), is a type of electrochemical cell where chemical energy is provided by two chemical components dissolved in liquids contained within the system and separated by a membrane. Because of chemical energy, ions (charged atoms or molecules) are generated and flow through the membrane. These produce an electric charge while both liquids circulate in their own respective space.

A flow battery may be used like a fuel cell (where spent fuel is extracted and new fuel is added to the system) or like a rechargeable battery (where an electric power source drives regeneration of fuel). While it has technical advantages over conventional rechargeable ones, such as potentially separable liquid tanks and near unlimited longevity, currents are comparatively less powerful and require more sophisticated electronics. Energy capacity is a function of the electrolyte volume (amount of liquid electrolyte), and power is a function of the surface area of electrodes.

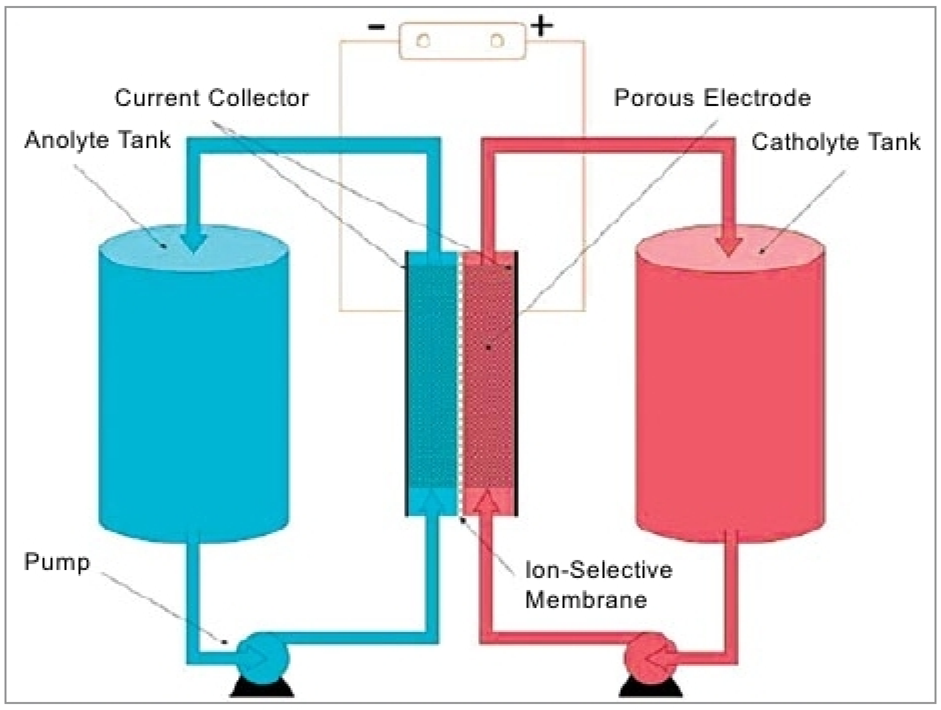

Fig. 3: Working of flow battery

Flow batteries are unique and the most practical for renewable energy to road transport due to the following reasons:

- Rapid and safe charging by electrolyte exchange
- Wheel-to-wheel efficiency four times better than hydrogen as fuel
- Operating costs equal to conventional diesel-fuelled vehicles
- Totally recycled fuel
- Zero emission at point of use

Motor controller. Control of an EV is not a simple task in that its operation is essentially time-variant (for example, operation parameters of an EV and road conditions are always varying). Therefore the controller should be designed to make the system robust and adaptive, improving the system on both dynamic- and steady-state performances.

Currently, the major limiting factor for the widespread use of EVs is the short running distance per battery charge. Hence, besides controlling the performance of the vehicles (that is, smooth driving for comfortable riding), significant efforts must be made for the energy management of batteries. However, from the viewpoint of electric and control engineering, EVs are advantageous over traditional vehicles with an internal combustion engine. Basic performance requirements of the EVs' motor drive system are high performance, low loss, high power density, low speed, high torque, a wide range of variable speed, strong overload capacity and good reliability.

Presently, brushed DC motor, brushless DC motor, AC induction motor, permanent magnet synchronous motor (PMSM) and switched reluctance motor (SRM) are the main types of motors used for EV driving. Selection of the motor for a specific EV is dependent on such factors as intention of the EV, ease of control and so on.

To control an EV, the objective is to control the torque of the driving machine. The throttle position and the brake are inputs to the control system, which is required to be fast-responsive and low-ripple. An EV requires that the driving electric machine has a wide range of speed regulation. To guarantee speed-up time, the electric machine must have large torque output under low speed and high overload capability. And to operate at high speed, the driving motor should have certain power output at high-speed operation.

Electric engine. The electric engine propels the electric car. An engine uses either AC or DC current. An AC engine, commonly used in electric cars, is lighter and less expensive than an engine that uses DC current. AC engines have fewer moving parts and are, hence, prone to fewer mechanical problems.

Regenerative braking. In a battery-powered EV, regenerative braking (also called regen) is the conversion of the vehicle's kinetic energy into chemical energy stored in the battery, where it can be used later to drive the vehicle. It is braking because it also serves to slow the vehicle. It is regenerative because energy is recaptured in the battery